

AVIRAL NIGAM

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Research Interests

AI safety through empirical LLM evaluation — chain-of-thought faithfulness, confidence calibration and honesty measurement, adversarial benchmark construction, scalable oversight, and black-box behavioral monitoring of agentic systems.

Education

Manipal Institute of Technology, MAHE

Manipal, India

B.Tech., Data Science and Engineering

Expected July 2028

- **CGPA:** $\frac{[[\text{ADD YOUR CGPA}]]}{10}$ (*Mitacs screens on a 70% minimum — do not submit without this*)
- **Relevant coursework:** Deep Learning, Statistical Modeling, Linear Algebra & Probability, Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems.
- **Honors:** Shortlisted, Algoverse AI Research Fellowship (Fall 2026 cohort).

Publications

Last Translation Benchmark (LTB) — *co-author*

Sept 2026

V. Zouhar, N. Bafna, ..., **A. Nigam**, et al. (community-authored, 170+ contributors). *Last Translation Benchmark*. arXiv:[2609.04173](https://arxiv.org/abs/2609.04173) [cs.CL], 2026.

- Multi-institution benchmark (ETH Zurich, Johns Hopkins, Charles University, UvA, KIT and others): 3,456 human-authored, peer-reviewed examples across 58+ languages spanning text, image, audio and video, each paired with handcrafted verification rules for objective scoring where automatic metrics saturate.
- Contributed **adversarial multimodal examples that broke state-of-the-art model performance** — direct evidence that headline benchmark scores do not survive targeted adversarial construction.
- Dataset: huggingface.co/datasets/zouhar/last-translation-benchmark | Leaderboard: last-translation-benchmark.vilda.net

AI Safety & Evaluation Research

FactTrace — Chain-of-Thought Faithfulness Monitor github.com/Aviral0gHub/facttrace

Aug 2026

- Claim-level factuality verifier for **agentic LLM reasoning traces**: decomposes multi-step reasoning into atomic claims, retrieves evidence via FAISS dense retrieval, and labels each claim supported, contradicted, or unverifiable using NLI entailment scoring — black-box, requiring no model internals.
- Improved claim-level accuracy from 78.5% (lexical baseline) to **87.7%** on a 65-claim curated benchmark by replacing regex heuristics with dense retrieval and RoBERTa-large-MNLI — a 9-point swing showing how much of a faithfulness result is an artifact of metric design.
- Found **unsupported claims concentrate in later reasoning steps**, precisely where a trace-based monitor is least likely to catch them; benchmark size and single-model scope documented as stated limitations.

CalibCheck — Calibration & Honesty Evaluation github.com/Aviral0gHub/calibcheck

Aug 2026

- Epistemic-evaluation pipeline measuring whether a model's *stated* confidence tracks its accuracy: **verbalized self-reported confidence vs. raw next-token probability**, scored by Expected Calibration Error, Brier score and reliability diagrams across 6 topic domains.
- Found Qwen2.5-0.5B-Instruct overconfident by 14.3 points on average (**ECE = 0.14**) over 60 questions, with domain-conditional miscalibration up to **57.5 points** on sports (30% accuracy vs. 87.5% stated confidence); small-*n* variance caveats reported explicitly.

Technical Skills

Evaluation & AI Safety Behavioral and black-box evals, calibration (ECE, Brier, reliability diagrams), claim decomposition, retrieval-augmented verification, NLI/entailment scoring, adversarial benchmark construction

ML & Models PyTorch, HuggingFace Transformers, Sentence-Transformers, scikit-learn, FAISS, metric & transfer learning; Qwen2.5-0.5B-Instruct, RoBERTa-large-MNLI, ResNet

Languages & Tooling Python, Java, Shell/Bash, Linux, Git, CUDA